The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Supplementary Material

Appendices and supplementary figures accompanying the article. Section, equation and figure numbers referenced here follow the main text; supplementary figures are numbered S1, S2, ...

A Notation and units

These neural-mass models are phenomenological: potentials are in mV, firing rates in s^{-1} ($\equiv$Hz), time constants in ms, and frequencies in Hz, unless noted; a "–" marks a dimensionless quantity. A few glyphs are reused across the four models (JR, LaNMM, NMM2/MPR, QIF) by long-standing convention; these are disambiguated by context and flagged below. In particular, A is the excitatory PSP *gain* in the JR/LaNMM synapses and nothing else; the applied-field modulation amplitude is ε in JR and LaNMM alike, and A_f in NMM2/QIF. These are not interchangeable units. In JR and LaNMM the field is a membrane polarization in mV entering the sigmoid argument, so ε is measured against the sigmoid voltage scale $1/\rho \approx 1.8$ mV, which is the same in both models. In NMM2/QIF the field enters the excitatory membrane current in the dimensionless MPR units of v_E, and A_f carries no volt scale at all. Likewise C is the JR connectivity constant and the NMM2 global synaptic gain C ($\equiv J$).

Symbol	**Meaning**	**Units**
Firing-rate nonlinearity (sigmoid)		
$\sigma(v)$	population firing-rate transfer function	s^{-1}
$\sigma'(v)$, $\sigma''(v)$	sigmoid slope and curvature; σ'' sets the demodulation gain	$\mathrm{s}^{-1}\mathrm{mV}^{-1,-2}$
e_0	half the maximal firing rate ($\sigma \to 2e_0$ as $v \to \infty$)	s^{-1}
v_0	sigmoid inflection (half-maximal) potential	mV
ρ	sigmoid slope parameter	mV^{-1}
v^*	operating point (field-free fixed point of v)	mV
$v = y_1 - y_2$	net pyramidal membrane potential (LFP proxy)	mV
PEIX	Population Excitation–Inhibition Index (a σ'-based brain-state index)	–
Single-column Jansen–Rit		
A, B	excitatory, inhibitory PSP gains	mV
a, b	excitatory, inhibitory synaptic rates (inverse time constants)	s^{-1}
y_1, y_2	excitatory, inhibitory post-synaptic potentials	mV
C, C_1-C_4	intracolumn connectivity constants	–
p	external input rate; JR bifurcation parameter	s^{-1}
Applied TI field and coupling		
$s(t)$, $e(t)$	applied TI field perturbation, AM or two-tone (JR; LaNMM)	mV
ε	field modulation amplitude, JR and LaNMM (a membrane polarization entering the sigmoid argument)	mV
A_f	field modulation amplitude, NMM2/QIF (in the dimensionless MPR units of v_E)	–
m	modulation depth	–
f_c, $\omega_c = 2\pi f_c$	carrier frequency	Hz, $\mathrm{rad\,s}^{-1}$
f_1, f_2	the two applied carrier tones ($f_c = (f_1+f_2)/2$)	Hz
Δf, $\Omega = 2\pi\Delta f$	envelope (beat) frequency, $\Delta f = f_2 - f_1$	Hz, $\mathrm{rad\,s}^{-1}$
$E(t)$, $\delta V(t) = \lambda E$	applied field; induced membrane perturbation	$\mathrm{V\,m}^{-1}$; mV
λ	lumped (polarization-length) field-to-membrane coupling	mV / ($\mathrm{V\,m}^{-1}$)
$\kappa(f_c)$	applied-field-to-polarization coupling	mV / ($\mathrm{V\,m}^{-1}$)
$H_m(\omega)$, τ_m	somatic membrane low-pass and its time constant	–; ms
τ	fast-element (node/AIS/terminal) time constant	ms
Network dynamics, response, and bifurcation		
$\lambda_\pm = -\gamma \pm i\omega_0$	leading Jacobian eigenvalue pair	s^{-1}
γ	damping of the critical mode ($\propto$ distance to the Hopf)	s^{-1}
ω_0, $f_0 = \omega_0/2\pi$	natural (Hopf) frequency	$\mathrm{rad\,s}^{-1}$, Hz
$\mathcal{Q} = \omega_0/2\gamma = \pi f_0 \tau$	resonance quality factor ($\tau = 1/\gamma$ the reverberation time)	–
$G(\Omega)$	network resonance gain (Lorentzian, $\propto 1/\gamma$ at ω_0)	–
A_Ω	demodulated response: lock-in amplitude of v at Ω	mV

Symbol	Meaning	Units
A_{open}	same lock-in, open loop (detector only, no recurrence); $A_\Omega/A_{\mathrm{open}}$ dimensionless	mV
I, Q	in-phase, quadrature lock-in components	mV
C^*, J^*	coupling at the Hopf (criticality)	–
Laminar model (LaNMM)		
P_1, P_2	deep (alpha) and superficial (gamma) pyramidal populations	–
$\mathrm{PV}, \mathrm{SS}, \mathrm{SST}$	parvalbumin, stellate, somatostatin interneurons	–
I_0	flat (DC) drive offset in $e(t)$	mV
Exact mean field (NMM2 / MPR) and QIF network		
r_E, r_I	population firing rates (E, I)	s^{-1}
v_E, v_I	mean membrane potentials (E, I), in dimensionless MPR units	–
η_j, $\bar{\eta}$	per-neuron excitability; mean excitability (= common drive, bifurcation param.)	–
Δ	Lorentzian half-width of η_j (excitability heterogeneity)	–
$\tau_E, \tau_I, \tau_a, \tau_g$	E, I membrane and AMPA, GABA synaptic time constants	ms
$C\,(\equiv J)$	global synaptic coupling (inter-population gain)	–
$A_{ee}, A_{ei}, A_{ie}, A_{ii}$	synaptic weight matrix $(1, 1, 1, 2)$	–
s_E, s_I	second-order synaptic activations; $\hat{K}_a, \hat{K}_g$ their operators	s^{-1}
V_j, V_{peak}	QIF neuron potential; spike/reset threshold	mV
N	neurons per population (QIF; $N = 8000$)	–
$A_\Omega/\langle r_E\rangle$	Δf modulation depth (envelope-locked timing metric)	–

Table 1: Symbols, meanings, and (nominal) units. Reused glyphs are disambiguated by context; see the head note on A and C.

B TI requires detection: the Stuart–Landau normal form amplifies but cannot demodulate

Amplification and detection are separable, and the universal Hopf normal form makes the separation sharp. The Stuart–Landau equation [1] is

$$\dot{z} = (a + i\omega_0)z - |z|^2 z + F(t), \qquad \gamma \equiv -a > 0 \;\; \text{(stable side)}, \tag{1}$$

driven by the TI field F. Expanding $z = \sum_k Z_k(t)e^{ik\omega_c t}$ in carrier harmonics with slowly varying envelopes, the field forces only $k = \pm 1$ and the cubic feeds index $k = l - p + q$. The odd-k subspace is therefore invariant and attracting: if every even Z_k vanishes, then for even k both sources vanish—$F_k = 0$, and $l - p + q$ cannot be even with l, p, q odd—leaving $\dot{Z}_k = [a + i(\omega_0 - k\omega_c)]Z_k$, which decays at rate $\gamma > 0$. So z lives only on odd carrier harmonics, to all orders in ε. The beat line is a *baseband* signal and would sit in $k = 0$; since $Z_0 \equiv 0$, the response at Δf is identically zero. An *even* nonlinearity instead produces a nonzero $k = 0$ term through $\cos^2 \omega_c t \to \frac{1}{2}$—the very line the odd cubic cannot make. Numerically ($a = -0.5$, $f_0 = 10$, $f_c = 100$ Hz, $\varepsilon = 0.5$), a square-law detector on the same field gives a Δf lock-in of 2.5×10^{-1} while $\mathrm{Re}\, z$ gives $\sim 2 \times 10^{-8}$, at the numerical floor. Driven instead directly in band, the same equation shows the forced $1/\gamma$ growth toward the Hopf.

The normal form thus responds to tACS but is blind to TI. Amplification is the model-independent near-Hopf susceptibility, shared with any normal-form resonator; detection is the model-specific even-order term the normal form lacks. A critical resonator alone cannot demodulate.

C Vocabulary and metrics

"Entrainment" is used in several senses, and our system can exhibit more than one, so we fix the vocabulary. We reserve *entrainment* for its dynamical-systems meaning, frequency/phase locking of a *self-sustained* oscillator to an external rhythm confined to an Arnold tongue [1], and distinguish it from two related read-outs. *(i) Forced response.* Below the Hopf the column does not oscillate on its own; the field elicits a driven response whose susceptibility we quantify by the lock-in amplitude of the read-out at the beat Δf (Eq. (19)), and there is no autonomous rhythm to entrain (J-curve, timing, tACS). *(ii) Frequency entrainment.* Above the Hopf the column free-runs at f_0; genuine entrainment means its dominant frequency is *pulled* to Δf. We test this with the dominant output frequency f_{out} (largest non-DC FFT peak) and the 1:1 criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz, mapping the Arnold tongue and locking range (Fig. S3); we use f_{out} rather than a phase-locking value [2] because, when an autonomous rhythm and a driven component coexist, instantaneous-phase estimates conflate them whereas the dominant-frequency test isolates true frequency capture. *(iii) Envelope locking / gating.* A periodic drive can instead modulate the *amplitude* of a persisting rhythm at the beat without capturing its frequency—phase–amplitude coupling [3, 4]. We quantify this by the Δf-locked component of the population rate (lock-in of r_E at Δf, equivalently the rate *folded* on the Δf beat phase—averaged over many beat cycles as a function of phase within a single cycle, so that everything not locked to Δf averages away), which isolates the driven envelope from the intrinsic rhythm. The distinction is load-bearing: the same word would otherwise denote a frequency capture (ii) and an amplitude gating (iii) that are dynamically different—so we label frequency capture *entrainment* and amplitude modulation of a persisting rhythm *gating*, reporting the metric appropriate to each (the above-Hopf QIF panel, with its $\sim$11 Hz detuning, is a case of gating, not entrainment).

The Arnold-tongue protocol for Fig. S3 places the column on the limit-cycle side (input p between the SNIC and the Hopf, free-running at $f_0(p)$) and drives it with $s(t) = \varepsilon \cos \Omega t$; the implementation, including the grid widths and the clipping check, is in the released code. One subtlety is worth stating: as onset is approached the cycle amplitude vanishes and the tongue merges into the forced-resonance bandwidth, so the supercritical locking range and the subcritical $1/\gamma$ susceptibility coincide there by construction—which is why responsiveness peaks at criticality on both sides.

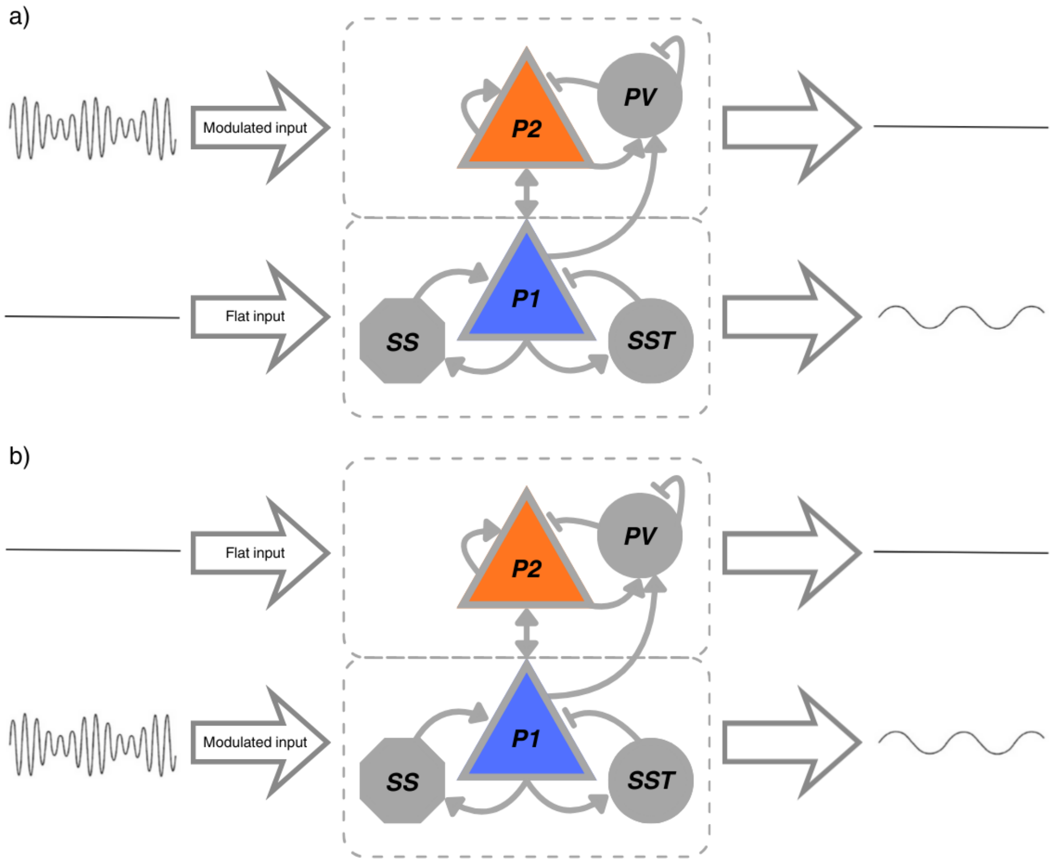

Figure S1: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the TI field drives P_2 while P_1 receives a flat drive; *(b)* the TI field drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the slow envelope is transferred to the column: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency beat component (right traces).

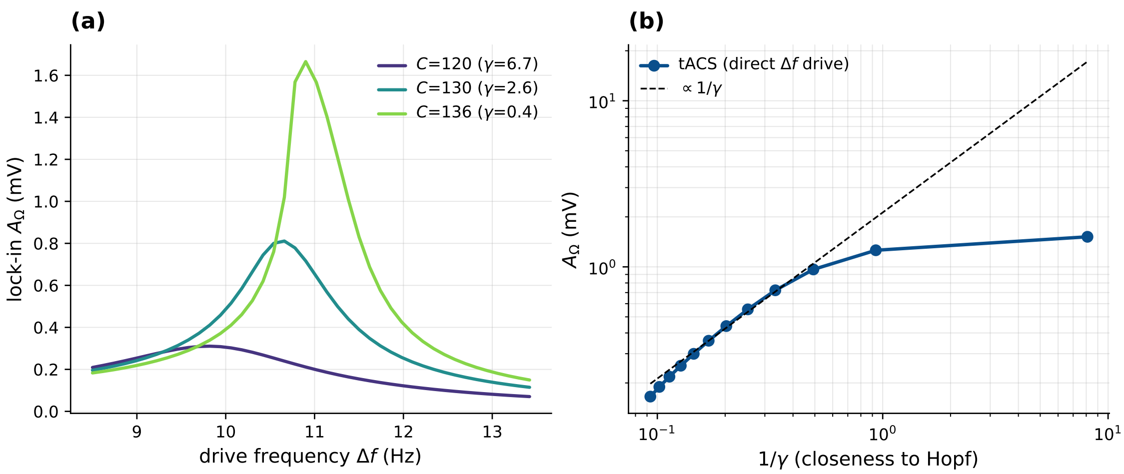

Figure S2: tACS shares the network amplifier (Jansen–Rit). A direct low-frequency (tACS) drive at Δf enters the column without a carrier, so it needs no demodulation, but feeds the same near-Hopf resonator. With the operating point pinned, sweeping the coupling $C\,(=J)$ toward the alpha Hopf: *(a)* the lock-in resonance at Δf sharpens and grows as $C \to C^*$; *(b)* the response grows as $1/\gamma$ (dashed) and saturates near criticality—the same network amplification as the demodulated TI drive (Fig. 9), but linear in the field rather than square-law. Throughout, the column is a stable focus ($\gamma > 0$).

D Supplementary figures

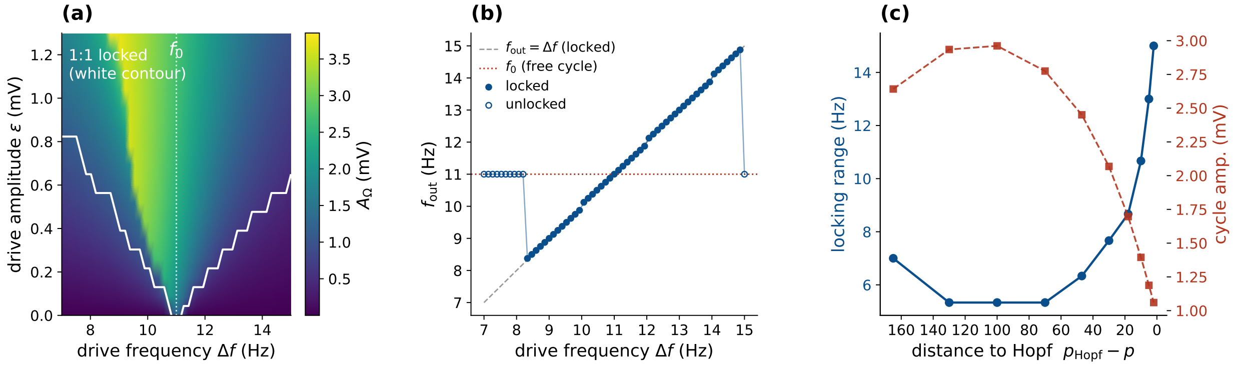

Figure S3: Above the Hopf: entrainment, not $1/\gamma$ amplification (Jansen–Rit). The column is set in its autonomous alpha limit cycle ($p = 250$, $f_0 \approx 11$ Hz) and driven by a direct in-band sinusoid. *(a)* The 1:1 Arnold tongue—where the dominant output frequency locks to Δf—is a wedge anchored at f_0 that widens with drive. *(b)* Inside the tongue the output follows Δf; outside, it falls back to f_0. *(c)* Toward the Hopf at fixed drive the locking range grows as the autonomous cycle amplitude vanishes (blue, left axis; red dashed, right axis). Above the Hopf amplification reads as tongue width rather than as the diverging $1/\gamma$ gain of the stable-focus side (Fig. 9): the column is most responsive near criticality on both sides.

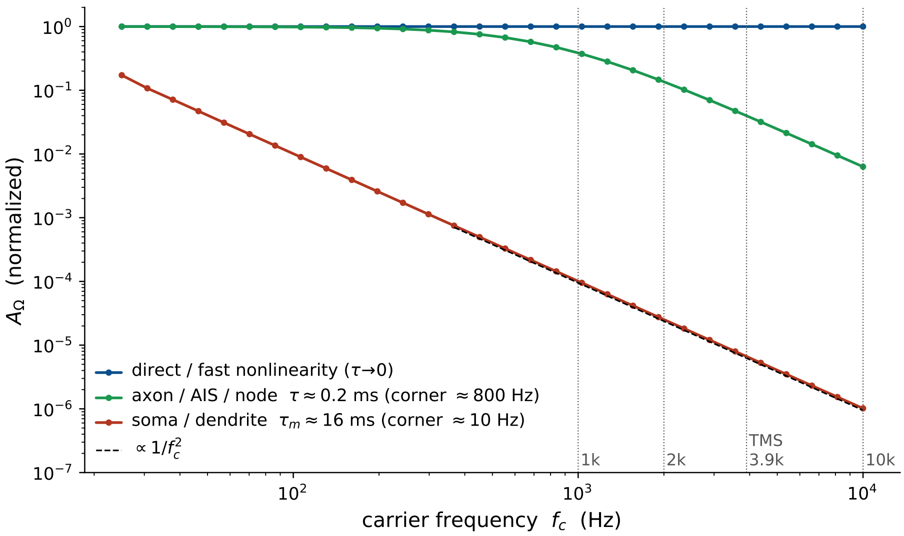

Figure S4: Somatic demodulation dies at kHz, but a fast element survives (Jansen–Rit, open-loop detector). Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide), killing kHz demodulation; a fast element (axon/AIS/node, $\tau \approx 0.2$ ms) retains an appreciable response into the kHz band ($\sim$0.14 at 2 kHz, $\sim$0.04 at the TMS peak 3.9 kHz). The same membrane filter attenuates TMS at the soma, which is why neither TMS nor TI should be thought of as driving the soma at kHz.

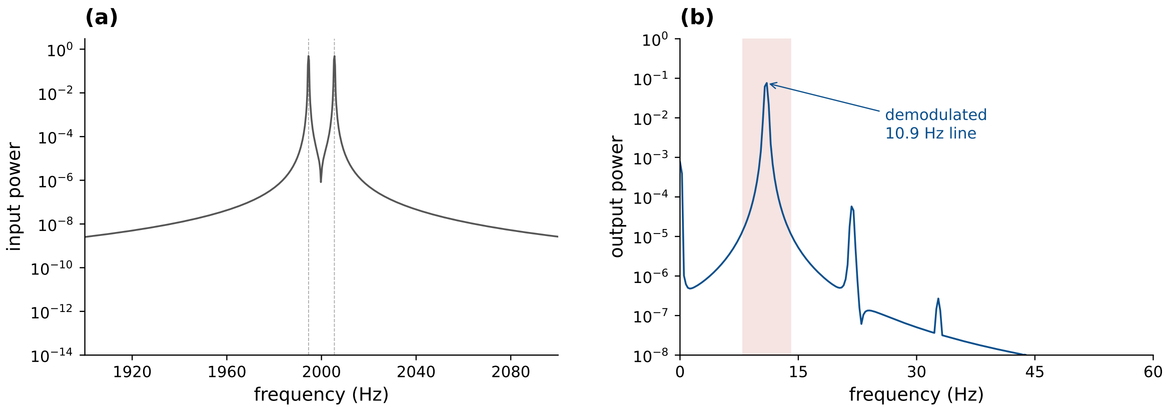

Figure S5: Genuine two-tone TI at a real kHz carrier (Jansen–Rit, fast-element route). Two equal tones at $f_1 = 1994.5$ and $f_2 = 2005.5$ Hz, beating at 10.9 Hz, pass through the fast-element low-pass ($\tau = 0.2$ ms) into the recurrent column. *(a)* The input spectrum has lines only at f_1 and f_2; power in the 8–14 Hz band is 2.2×10^{-13}, zero to numerical precision. *(b)* The output carries a clean demodulated line at 11.0 Hz. This is the kHz claim on the physical waveform rather than on the AM surrogate.

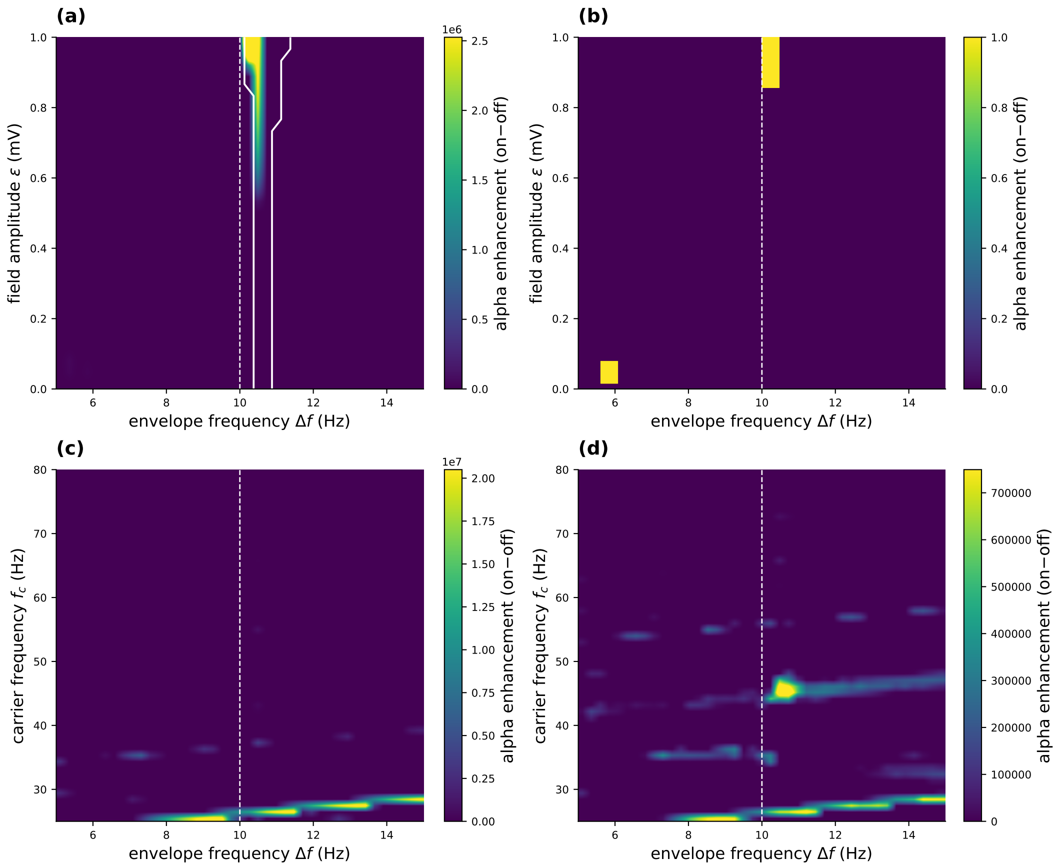

Figure S6: LaNMM Arnold tongues: two-tone TI field delivered directly to the deep P_1 (alpha) loop. As Fig. 7 but with the field on P_1. The deep loop is captured by a much weaker field than the superficial one, so the amplitude range is $\varepsilon \leq 1$ mV here against 3 mV there; above ≈2 mV the drive captures essentially the whole Δf sweep and leaves the perturbative regime the theory describes. Colour: stimulation-induced alpha-band enhancement over the $\varepsilon = 0$ baseline. *(a,b)* P_1 and P_2 enhancement vs. ε and Δf; the white contour in *(a)* outlines the 1:1 frequency-captured set, $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix C), computed from the dominant output frequency because band power alone cannot separate capture from a forced response. Its width grows from 0.5 Hz at $\varepsilon \to 0$ to 1.25 Hz at $\varepsilon = 1$ mV. *(c,d)* Enhancement vs. carrier frequency and Δf at $\varepsilon = 0.5$ mV. P_1 responds across a *broad* carrier band—a vertical ridge at $\Delta f \approx 10$ Hz, the direct-coupling limit of Eq. (10). Read with the lock-in at Δf the response is retained at every carrier from 25 to 80 Hz (median 25% of peak), in contrast to the ∼7% collapse off resonance under superficial drive.

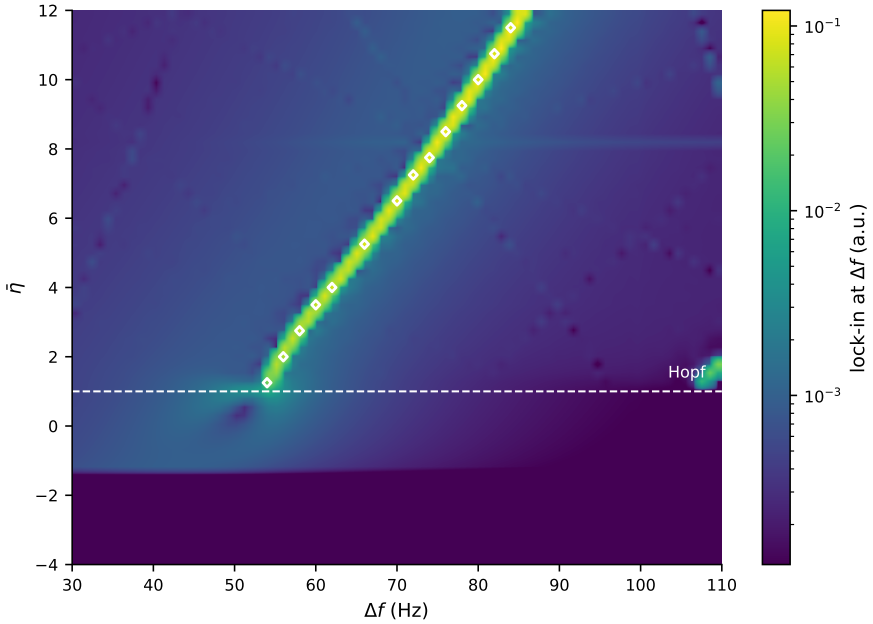

Figure S7: NMM2 PING resonance map (exact mean field). Demodulated response over the (Δf, background drive $\bar{\eta}$) plane, log color scale. The ridge follows the gamma natural frequency, and its diagonal slope—f_0 rising with $\bar{\eta}$—reflects the state-dependent transfer of NMM2, absent in the static-sigmoid models. White markers outline the 1:1 captured set (Appendix C), drawn only above the Hopf (dashed line) where there is an autonomous rhythm to capture; below it the same color field is a forced response. Lock-in amplitude cannot separate the two, since both place power at Δf.

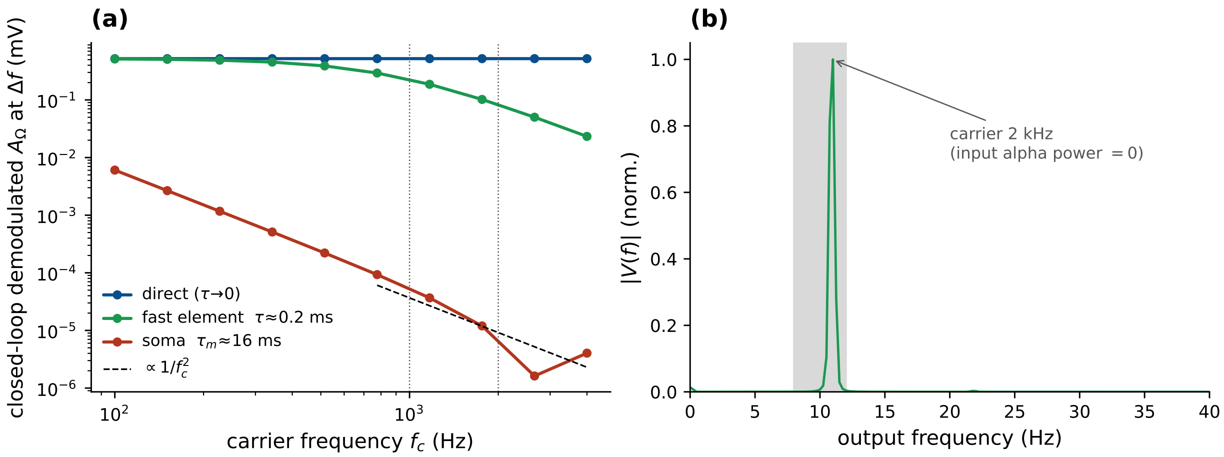

Figure S8: Demodulation at a genuine kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three coupling routes: direct ($\tau \to 0$, carrier-independent), fast element (axon/AIS/node, $\tau \approx 0.2$ ms; survives into the kHz band), and soma ($\tau_m \approx 16$ ms; collapses as $1/f_c^2$). *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: a clean alpha line at f_0, synthesized by the column although the input carries no alpha power.

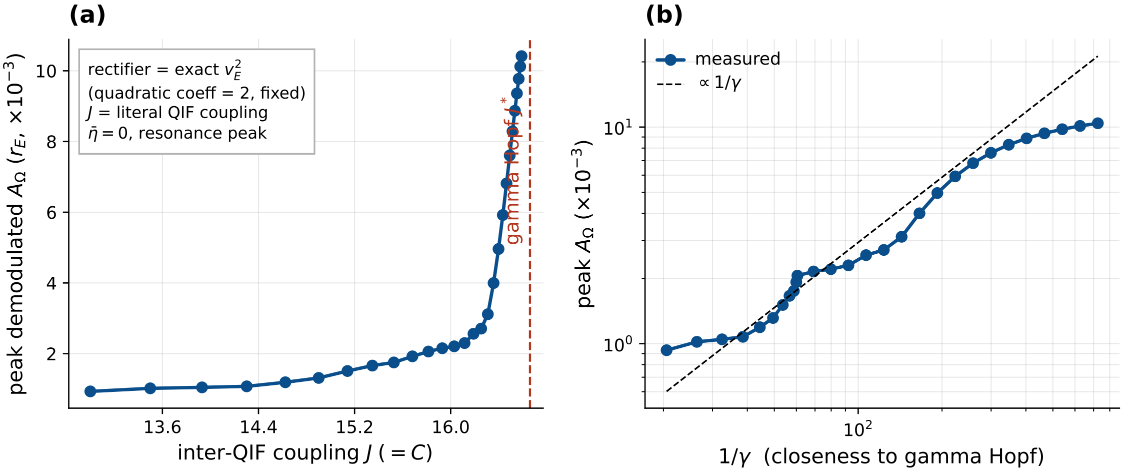

Figure S9: Coupling controls the amplification in the exact mean field (NMM2 PING, gamma). $J\,(=C)$ is the literal inter-QIF synaptic weight; the rectifier is the exact v_E^2 (fixed quadratic coefficient), so no operating-point pinning is needed. *(a)* The peak demodulated response of r_E grows as $J \to J^*$ (gamma Hopf, $\bar{\eta} = 0$; stable focus, $\gamma > 0$). *(b)* The gain grows as $1/\gamma$ (dashed), saturating near criticality—the JR law, now with coupling and rectifier both derived.

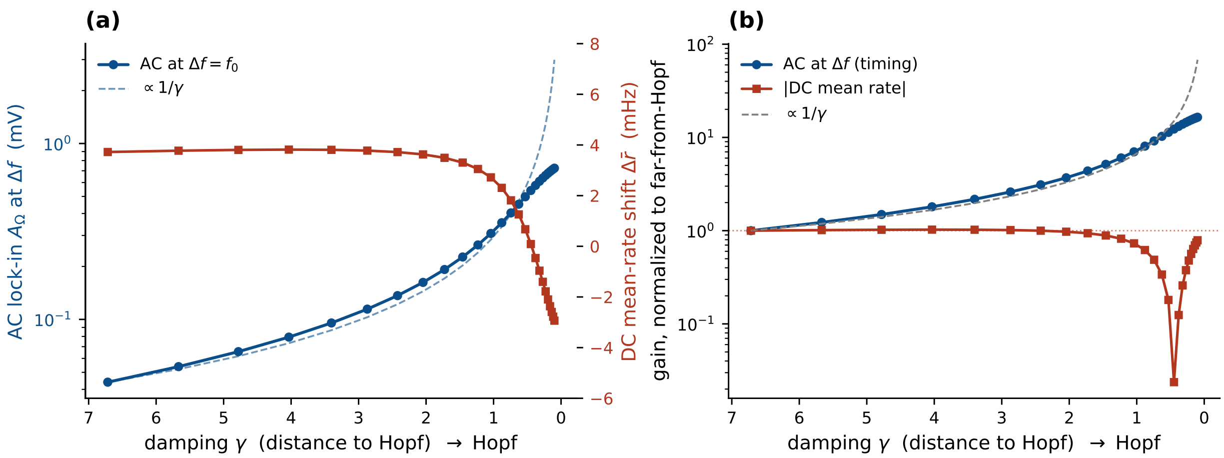

Figure S10: Timing, not rate (Jansen–Rit, detection pinned). As the column is driven toward the Hopf (from the stable side, $\gamma > 0$) through the coupling at fixed $\sigma''(v^*)$, *(a)* the AC lock-in at the resonant beat $\Delta f = f_0$ (blue) is resonantly amplified and tracks $1/\gamma$ (dashed), while the DC shift of the mean firing rate (red) stays flat at a few mHz. *(b)* Normalized to the far-from-Hopf value (log scale): the AC gain follows $1/\gamma$ and saturates near criticality, whereas the DC gain stays $\sim$1. The envelope locks timing, not rate—the population analog of the spike-timing-without-rate finding of [5].

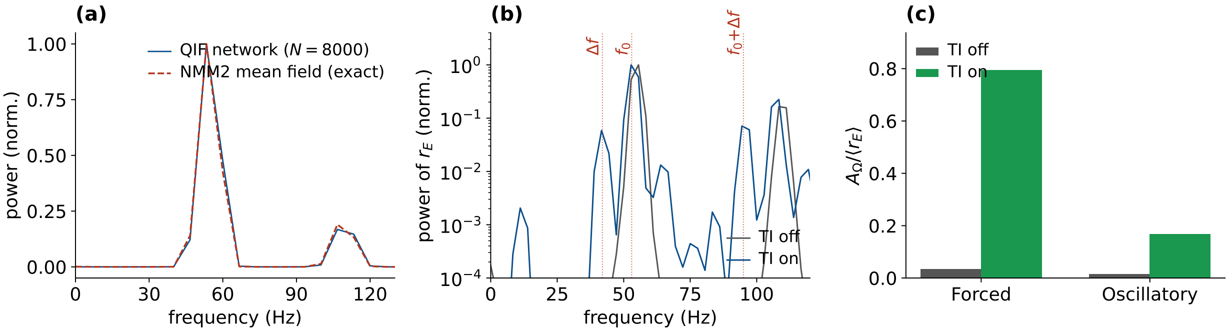

Figure S11: Timing, not rate, in the QIF network. *(a)* Field-free power spectrum of $r_E(t)$: the spiking network ($N = 8000$) matches the exact NMM2 mean field in both gamma frequency ($\sim$53 Hz) and mean rate. *(b)* Intermodulation above the Hopf (TI off vs. on, log power): the nonlinearity mixes the imposed $\Delta f = 42$ Hz beat with the intrinsic 53 Hz gamma, raising the direct Δf line and the sum sideband near 95 Hz, both absent with TI off. This is the sideband signature of gating rather than entrainment. *(c)* The Δf modulation depth $A_\Omega/\langle r_E\rangle$, off vs. on: 0.01–0.03 rising to 0.17–0.79, while the mean rate stays nearly flat ($\times$1.07–1.11). Modulation depth is reported rather than the off$\to$on lock-in ratio, whose denominator sits at the near-zero field-free floor.

References

[1] Arkady Pikovsky, Michael Rosenblum, and Jürgen Kurths. *Synchronization: A Universal Concept in Nonlinear Sciences.* Cambridge University Press, 2001.

[2] Jean-Philippe Lachaux, Eugenio Rodriguez, Jacques Martinerie, and Francisco J. Varela. Measuring phase synchrony in brain signals. *Human Brain Mapping*, 8(4):194–208, 1999.

[3] Ryan T. Canolty and Robert T. Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010.

[4] Adriano B. L. Tort, Robert Komorowski, Howard Eichenbaum, and Nancy Kopell. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2):1195–1210, 2010.

[5] Pedro G. Vieira, Matthew R. Krause, and Christopher C. Pack. Temporal interference stimulation disrupts spike timing in the primate brain. *Nature Communications*, 15:4558, 2024.